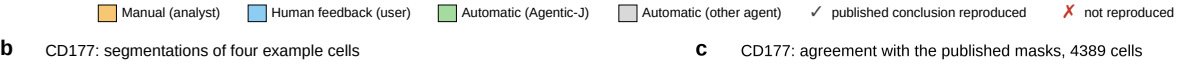
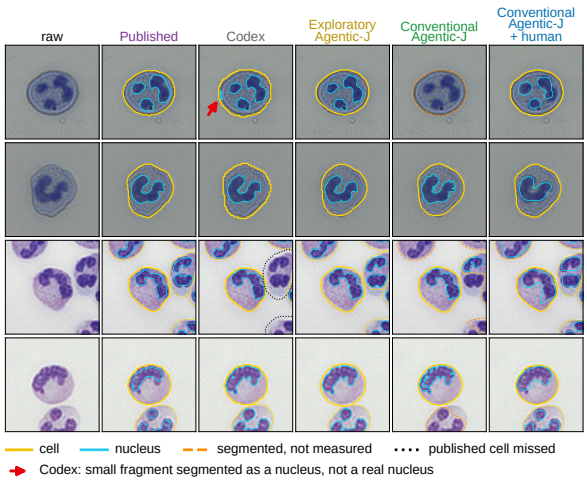
a CD177 neutrophil morphology

Do CD177+ and CD177− neutrophils differ in cell and nucleus morphology?

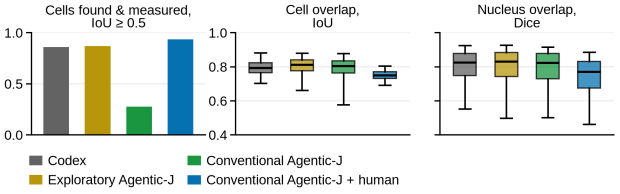
	Cell segmentation	Nucleus segmentation	QC & post-processing	Features	Statistics	Time to result	Scientific verdict
Published	Cellpose (default)	ilastik pixel classifier	dilation, crop, size filter	25 hand-picked	cell-level Mann–Whitney	weeks	no meaningful difference
Codex CLI	Cellpose cyto3 at half resolution	erosion/threshold within cell	shape QC	30 features	donor-level Wilcoxon, FDR	13 min	✓ 0/30 features significant
Exploratory Agentic-J	Cellpose cyto3	Otsu threshold within cell	border, size, solidity filter	25 features	donor-level Wilcoxon, FDR	35 min	✗ 5/25 significant, 3/3 runs
Conventional Agentic-J	Cellpose cyto3 on optical density	threshold within cell	size/shape QC	32 features	donor-level permutation, FDR	67 min	✓ 0/32 significant, 3/3 runs
Conventional Agentic-J + human	Cellpose-SAM, chosen by the user	micro-sam fine-tuned on 11 hand-annotated tiles	lobe-to-cell assignment	11 features	image-level Mann–Whitney, FDR	488 min	✓ 0/11 features significant



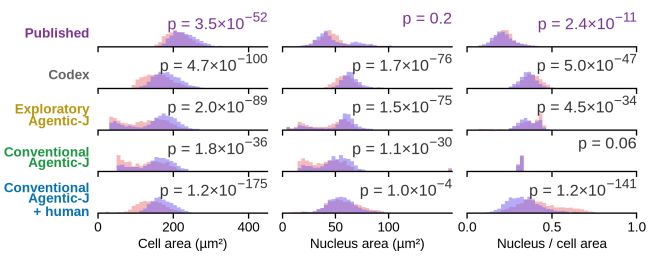
b CD177: segmentations of four example cells



c CD177: agreement with the published masks, 4389 cells



d CD177: per-cell distributions vs published data



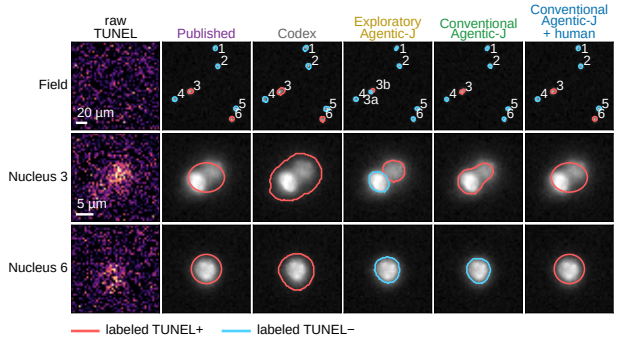
e Leishmania infection

Published finding: L. major proliferates more in apoptotic, TUNEL+, host macrophages than in TUNEL− ones.

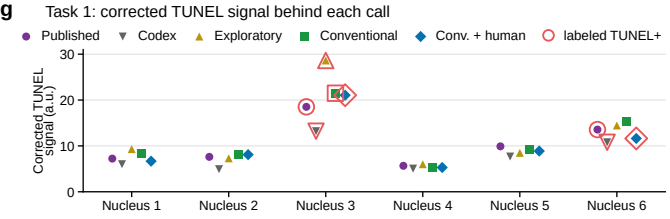
Task 1: decide which of 6 host nuclei are TUNEL+. Task 2: compare the proliferation index of 15 parasites between TUNEL+ and TUNEL− hosts.

	Registration, live → fixed	Host / nucleus segmentation	TUNEL call	Parasite localisation	Measurement & statistics	Time T1 / T2	Task 1 TUNEL+ and TUNEL− nuclei 6 nuclei, fixed frame	Task 2 proliferation in TUNEL+ hosts 15 parasites, last live frame
Published	by hand	hand-drawn ROIs	by inspection	15 ROIs + 3 backgrounds each, by hand	ROI Manager → Excel; Mann–Whitney	not recorded	TUNEL+: nuclei 3, 6 TUNEL−: nuclei 1, 2, 4, 5	higher proliferation in TUNEL+ hosts 8 TUNEL+ vs 7 TUNEL− parasites
Codex CLI	SIFT + RANSAC, ECC-refined to 1.6 px	T1 DAPI threshold; T2 Cellpose cyto3	largest-gap cutoff	red-channel peaks + water-shed: 10 kept	annulus background; no test	18 min, both tasks	TUNEL+: nuclei 3, 6 TUNEL−: nuclei 1, 2, 4, 5	✓ higher proliferation in TUNEL+ hosts Δ median 1.68; 10 of 15 parasites
Exploratory Agentic-J	TurboReg rigid on brightfield	T1 StarDist; T2 Cellpose cyto3, 7 hosts	largest-gap cutoff	C2 + C3 spot evidence: 22 kept	annulus background; no test	22 / 69 min	TUNEL+: nucleus 3, split in two TUNEL−: nuclei 1, 2, 4, 5, 6	✗ not testable: no parasite found 22 false detections, no host
Conventional Agentic-J	TurboReg / StackReg rigid	T1 DoG + water-shed; T2 Cellpose cyto3, 6 hosts	2-component GMM	micro-sam vit_b_lm: 105 → 5 kept	annulus background; no test	55 / 85 min	TUNEL+: nucleus 3 TUNEL−: nuclei 1, 2, 4, 5, 6	✗ not testable: no TUNEL+ host found 3 of 15 parasites, 2 false
Conventional Agentic-J + human	lab ROIs given, origin translated	lab ROIs, given	3×SD rule overruled; user review	lab ROIs, 3 backgrounds user-approved	native 16-bit; Mann–Whitney	35 / 45 min	TUNEL+: nuclei 3, 6 TUNEL−: nuclei 1, 2, 4, 5	✓ higher proliferation in TUNEL+ hosts Δ median 2.36 (published data 2.36); 15/15

f Task 1: TUNEL+ and TUNEL− calls on the six host nuclei



g Task 1: corrected TUNEL signal behind each call



h Tasks 1 and 2: counts compared with the published analysis

		Codex CLI	Exploratory Agentic-J	Conventional Agentic-J	Conventional Agentic-J + human
Task 1	TUNEL calls agree	6 / 6	5 / 6	5 / 6	6 / 6
Task 2	Lab parasites found	10 / 15	0 / 15	3 / 15	15 / 15
	Host group correct	10 / 10	–	2 / 3	15 / 15
	Index within 0.1 of lab	4 / 10	–	1 / 3	14 / 15
	False detections	0	22	2	0

i Task 2: parasite proliferation in TUNEL+ vs TUNEL− host macrophages

